STAT6611

- Lecture 06 (R, Sept 10)
- Homework 1 assigned
  - practice on g.i.2 matrix results
  - due F, Sept 18 (F next week) by midnight in gradescope/canvas
- My office hours: TW 12:30-1:30 PM
  & R 1-2 PM

TA Ben: R 9:30-10:30 AM
in Snedecor 2207

2.5 The Conjugate Normal Equations

The least squares estimate of an estimable function $\underline{\lambda}'\underline{\beta}$ can be obtained by computing a solution $\underline{\hat{\beta}}$ to the normal equations and by then forming $\underline{\lambda}'\underline{\hat{\beta}}$, which is unique and can be written as $\underline{\lambda}'(X'X)^{-}X'\underline{y}$.

Alternatively, we can compute a solution $\underline{r}$ to the conjugate normal equations

$$X'X\underline{r} = \underline{\lambda}$$

and obtain the least squares estimate of $\underline{\lambda}'\underline{\beta}$ as $\underline{r}'X'\underline{y}$.

Theorem 2.7. *Under Models II—VII, $\underline{\lambda}'\underline{\beta}$ is estimable if and only if there exists an $\underline{r}$ such that $X'X\underline{r} = \underline{\lambda}$; that is, if and only if the conjugate normal equations are consistent.*

Corollary. Under Models II—VII, $\underline{\lambda}'\underline{\beta}$ is estimable if and only if $X'X(X'X)^{-}\underline{\lambda} = \underline{\lambda}$.

Theorem 2.8. *Under Models II—VII, the least squares estimate of an estimable function* $\underline{\lambda}'\underline{\beta}$ *is equal to* $\underline{r}'X'\underline{y}$*, where* $\underline{r}$ *is any solution to* $X'X\underline{r} = \underline{\lambda}$*.*

For any solution to $X'X\underline{r} = \underline{\lambda}$, we have

$$\underline{r}'X'\underline{y} = \underline{r}'X'P_X\underline{Y} = \underline{r}'X'(X(X'X)^{-}X')\underline{Y}$$

$$= \underline{r}'X'X\left((X'X)^{-}X'\underline{Y}\right)$$

$\underline{r}'X'X = \underline{\lambda}'$; $(X'X)^{-}X'\underline{Y}$ = solution $\hat{\underline{\beta}}$ to NEs

$$= \underline{\lambda}'\hat{\underline{\beta}}$$ ← unique least squares estimator of $\underline{\lambda}'\underline{\beta}$

properties

2.6 Expectations/Variances/Covariances of Least Squares Estimators

Suppose that $\underline{\lambda}' \in \mathcal{R}(X)$ so that $\underline{\lambda}'\underline{\beta}$ is estimable under Models II—VII. Taking $\underline{r}$ to be any solution to the conjugate normal equations

$$X'X\underline{r} = \underline{\lambda},$$

the ordinary least squares estimator of $\underline{\lambda}'\underline{\beta}$ is then $\underline{a}'\underline{y}$, where $\underline{a}' = \underline{r}'X'$ and also

$$\underline{a}'X = \underline{r}'X'X = \underline{\lambda}'.$$

Under Models II—VII, the ordinary least squares estimator of $\underline{\lambda}'\underline{\beta}$ is an unbiased estimator of the estimable function $\underline{\lambda}'\underline{\beta}$.

For two random vectors $\underline{x} = (x_1, \ldots, x_m)'$ and $\underline{z} = (z_1, \ldots, z_n)'$, we write $\text{cov}(\underline{x}, \underline{z})$ for the $m \times n$ matrix whose (i, j)th element is the covariance between x_i and z_j. That is,

$$\text{cov}(\underline{x}, \underline{z}) = \begin{bmatrix} \text{cov}(x_1, z_1) & \cdots & \text{cov}(x_1, z_n) \\ \vdots & \vdots & \vdots \\ \text{cov}(x_m, z_1) & \cdots & \text{cov}(x_m, z_n) \end{bmatrix}.$$

The matrix $\text{Var}(\underline{x}) = \text{cov}(\underline{x}, \underline{x})$ is called the variance-covariance matrix, or simply the covariance matrix, of $\underline{x}$.

Lemma 2.4. *If $\underline{x}$ is a random vector and L and M are matrices of constants having the same number of rows as $\underline{x}$ has elements, then*

$$\text{cov}(L'\underline{x}, M'\underline{x}) = L'\Sigma M, \text{ where } \Sigma = \text{Var}(\underline{x}).$$

Corollary. $\text{Var}(L'\underline{x}) = L'\Sigma L$, where $\Sigma = \text{Var}(\underline{x})$.

Definition. For a matrix of constants Λ having p rows, the vector $\Lambda'\underline{\beta}$ is said to be estimable if each of its elements is estimable.

Suppose that $\Lambda_1'\underline{\beta}$ and $\Lambda_2'\underline{\beta}$ (where Λ_1 and Λ_2 are matrices of constants having p rows each) are both estimable. Let $\widehat{\underline{\beta}}$ represent any solution to the normal equations, and take R_1 and R_2 to be any two matrices satisfying

$$X'XR_1 = \Lambda_1 \quad \text{and} \quad X'XR_2 = \Lambda_2.$$

The least squares estimators of $\Lambda_1'\underline{\beta}$ and $\Lambda_2'\underline{\beta}$, that is, the vectors whose elements are the ordinary least squares estimators of the corresponding elements of $\Lambda_1'\underline{\beta}$ and $\Lambda_2'\underline{\beta}$, are given by

$$\Lambda_1'\widehat{\underline{\beta}} = R_1'X'\underline{y} \quad \text{and} \quad \Lambda_2'\widehat{\underline{\beta}} = R_2'X'\underline{y}.$$

Under Model VII, by Lemma 1.4 and its Corollary 1 (ii),

$$\begin{aligned}
\operatorname{cov}(\Lambda_1'\widehat{\underline{\beta}}, \Lambda_2'\widehat{\underline{\beta}}) &= \operatorname{cov}(R_1'X'\underline{y}, R_2'X'\underline{y}) \\
&= R_1'X'VXR_2 \\
&= R_1'[X'X(X'X)^-X']V[X(X'X)^-X'X]R_2 \\
&= (R_1'X'X)(X'X)^-X'VX(X'X)^-(X'XR_2) \\
&= \Lambda_1'(X'X)^-X'VX(X'X)^-\Lambda_2,
\end{aligned}$$

and, in particular,

$$\operatorname{Var}(\Lambda_1'\widehat{\underline{\beta}}) = \Lambda_1'(X'X)^-X'VX(X'X)^-\Lambda_1.$$

Under Model II (the Gauss-Markov model), which can be regarded as a special case of Model VII,

$$\operatorname{cov}(\Lambda_1'\widehat{\underline{\beta}}, \Lambda_2'\widehat{\underline{\beta}}) = R_1'X'(\sigma^2 I)XR_2 = \sigma^2 R_1'X'XR_2 = \sigma^2 R_1'[X'X(X'X)^-X'X]R_2$$
$$= \sigma^2(R_1'X'X)(X'X)^-(X'XR_2) = \sigma^2\Lambda_1'(X'X)^-\Lambda_2,$$

and, in particular,

$$\operatorname{Var}(\Lambda_1'\widehat{\underline{\beta}}) = \sigma^2\Lambda_1'(X'X)^-\Lambda_1.$$

2.7 The Gauss-Markov Theorem

Under Models II—VII, the (ordinary) least squares estimator of an estimable function $\underline{\lambda}'\underline{\beta}$ is unbiased. Moreover, under Model II, the least squares estimator of an estimable function $\underline{\lambda}'\underline{\beta}$ is the best (minimum variance) linear unbiased estimator (BLUE) as described by the following Gauss-Markov theorem.

Theorem 2.9. (**The Gauss-Markov Theorem**) *Under Models II—VII, if $\underline{\lambda}'\underline{\beta}$ is estimable and $\widehat{\underline{\beta}}$ is any solution to the normal equations, then $\underline{\lambda}'\widehat{\underline{\beta}}$ is a linear unbiased estimator of $\underline{\lambda}'\underline{\beta}$. Under Model II, the variance of $\underline{\lambda}'\widehat{\underline{\beta}}$ is uniformly less than that of any other linear unbiased estimator of $\underline{\lambda}'\underline{\beta}$, that is, $\underline{\lambda}'\widehat{\underline{\beta}}$ is the best (minimum variance) linear unbiased estimator (BLUE) of $\underline{\lambda}'\underline{\beta}$.*

Proof. If $c + \underline{a}'\underline{y}$ is any linear unbiased estimator of $\underline{\lambda}'\underline{\beta}$, then $c = 0$, $\underline{\lambda}' = \underline{a}'X$ (Lemma 2.1); and

$$\mathrm{Var}(c + \underline{a}'\underline{y}) = \mathrm{Var}(\underline{a}'\underline{y}) = \mathrm{Var}[\underline{\lambda}'\widehat{\underline{\beta}} + (\underline{a}'\underline{y} - \underline{\lambda}'\widehat{\underline{\beta}})]$$

$$= \mathrm{Var}(\underline{\lambda}'\widehat{\underline{\beta}}) + \mathrm{Var}(\underline{a}'\underline{y} - \underline{\lambda}'\widehat{\underline{\beta}}) + 2\mathrm{cov}(\underline{\lambda}'\widehat{\underline{\beta}}, \underline{a}'\underline{y} - \underline{\lambda}'\widehat{\underline{\beta}}).$$

Finally, if $\text{Var}(c + \underline{a}'\underline{y}) = \text{Var}(\underline{\lambda}'\widehat{\underline{\beta}})$ ever holds, it must be that $\text{Var}(\underline{a}'\underline{y} - \underline{\lambda}'\widehat{\underline{\beta}}) = 0$ or

$$\text{Var}(\underline{a}'\underline{y} - \underline{\lambda}'\widehat{\underline{\beta}}) = \text{Var}(\underline{a}'\underline{y} - \underline{r}'X'\underline{y}) = \text{Var}[(\underline{a} - X\underline{r})'\underline{y}]$$

$$= (\underline{a} - X\underline{r})'(\sigma^2 I)(\underline{a} - X\underline{r}) = \sigma^2(\underline{a} - X\underline{r})'(\underline{a} - X\underline{r}) = 0 \iff \underline{a} - X\underline{r} = \underline{0}$$

$$\iff \underline{a}' = \underline{r}'X' \iff \underline{a}'\underline{y} = \underline{\lambda}'\widehat{\underline{\beta}} \text{ for all } \underline{y} \iff c + \underline{a}'\underline{y} = \underline{\lambda}'\widehat{\underline{\beta}} \text{ for all } \underline{y}.$$

That is, $\underline{\lambda}'\widehat{\underline{\beta}}$ is the unique best linear unbiased estimator of $\underline{\lambda}'\underline{\beta}$.

Suppose now that we wish to simultaneously estimate k estimable functions of $\underline{\beta}$.

- Choose the $p \times k$ matrix Λ in such a way that $\Lambda'\underline{\beta}$ is the vector whose elements are the k estimable functions.
- Let $\widehat{\underline{\beta}}$ be any solution to the normal equations.
- Then, the estimator $\Lambda'\widehat{\underline{\beta}}$ is one unbiased estimator of $\Lambda'\underline{\beta}$.

We might ask: In what sense (or senses) is $\Lambda'\widehat{\underline{\beta}}$ a "good" estimator?

To answer this, we consider some matrix properties regarding the variance of $\Lambda'\widehat{\underline{\beta}}$.

We'll first introduce some definitions.

Definitions. If $\underline{x}$ is a column vector of r real variables and A is an $r \times r$ matrix of constants, then we refer to the function

$$q(\underline{x}) = \underline{x}' A \underline{x}$$

as a quadratic form, where quadratic form $q(\underline{x})$ is said to be

nonnegative definite if $q(\underline{x}) \geq 0$ for all real vectors $\underline{x}$;

positive definite if $q(\underline{x}) > 0$ for all nonzero real vectors $\underline{x}$;

positive semidefinite if it is nonnegative definite but not positive definite (i.e., $q(\underline{x}) \geq 0$ with equality holding for at least one nonzero real vector $\underline{x}$)

A symmetric square matrix A is nonnegative definite, positive definite, positive semidefinite according to whether quadratic form $\underline{x}' A \underline{x}$ is nonnegative definite, positive definite, or positive semidefinite, respectively.

Theorem 2.10. *Under Model II, suppose that $\Lambda'\underline{\beta}$ is any k-dimensional estimable vector and that $\underline{c} + A'\underline{y}$ is any vector of linear unbiased estimators of the elements of $\Lambda'\underline{\beta}$. Then, the matrix $\mathrm{Var}(\underline{c} + A'\underline{y}) - \mathrm{Var}(\Lambda'\widehat{\underline{\beta}})$ is nonnegative definite, where $\widehat{\underline{\beta}}$ is any solution to the normal equations.*

Proof. For any $k \times 1$ vector $\underline{\ell}$, by the corollary to Lemma 2.4,

$$\underline{\ell}'[\mathrm{Var}(\underline{c}+A'\underline{y}) - \mathrm{Var}(\Lambda'\widehat{\underline{\beta}})]\underline{\ell} = \underline{\ell}'\mathrm{Var}(\underline{c}+A'\underline{y})\underline{\ell} - \underline{\ell}'\mathrm{Var}(\Lambda'\widehat{\underline{\beta}})\underline{\ell} = \mathrm{Var}[\underline{\ell}'(\underline{c}+A'\underline{y})] - \mathrm{Var}(\underline{\ell}'\Lambda'\widehat{\underline{\beta}}).$$

By Cor 2/Theorem 2.1, $\tilde{\underline{\lambda}}'\underline{\beta}$ for $\tilde{\underline{\lambda}}' \equiv \underline{\ell}'\Lambda'$ is estimable with least squares estimator as $\tilde{\underline{\lambda}}'\widehat{\underline{\beta}}$. Since $\underline{\ell}'(\underline{c} + A'\underline{y})$ is a linear unbiased estimator of $\tilde{\underline{\lambda}}'\underline{\beta} = (\underline{\ell}'\Lambda')\underline{\beta}$,